# Software Requirement Specification

## RailVision: Ai-Based Real Time Train Crowd Detection System

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**Feb,04,2024**

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# 1. Introduction

**RailVision** is an intelligent **train crowd detection system** designed to monitor passenger density inside train compartments using computer vision and deep learning techniques. The system uses a mobile phone camera instead of traditional CCTV cameras to reduce installation cost and make the system accessible and scalable.

The captured video is processed in real time to detect people, estimate crowd density, and display live visual information to passengers and railway authorities. This helps in preventing overcrowding, improving passenger comfort, and supporting operational decision-making.

## 1.1 Purpose:

This document specifies the functional and non-functional requirements for **RailVision: Real-Time Crowd Detection System**. The product is a vision-based system designed to monitor and report real-time crowd levels and occupancy in train compartments.

## 1.2 Scope :

The RailVision: AI-Based Train Crowd Detection & Monitoring System aims to overcome the limitations of traditional CCTV-based surveillance systems by implementing a low-cost, mobile-based, real-time, and AI-driven crowd analytics framework. The system focuses on accurately detecting passenger presence, estimating crowd density, and providing real-time visual feedback to both passengers and railway authorities to improve safety and travel experience.

### 1.2.1 In-Scope Functions (Deliverables):

The system will encompass the following key functionalities:

**Mobile Camera-Based Data Acquisition:**

Unlike conventional systems that depend on fixed CCTV infrastructure, this project will utilize a smartphone camera as the primary video input device to reduce deployment and maintenance costs.

**Real-Time Video Streaming & Processing:**

The system will continuously capture and process live video streams from the mobile camera, enabling real-time monitoring of train compartments.

**AI-Based Person Detection:**

The system will use deep learning-based computer vision models (such as YOLO/CNN) to detect and identify human presence in each video frame.

**Crowd Density Estimation:**

The system will calculate passenger density based on the number of detected individuals and classify crowd levels into categories such as:

Low

Medium

High

**Real-Time Passenger Visualization:**

The system will provide real-time visual feedback to passengers through a mobile or web interface, allowing them to view crowd conditions before entering a compartment.

**Data Storage & Analytics**

The system will store crowd statistics for historical analysis, reporting, and system performance

evaluation.

## 1.2.2 Out-of-Scope (Exclusions):

To ensure the project remains feasible within the academic timeline, the following features are excluded:

**Physical Infrastructure Control:**

The system will not control train operations such as door mechanisms, train scheduling, or coach allocation.

Biometric Identification

The system will not perform face recognition or identity tracking; only anonymous people counting is supported.

Automated Crowd Management

The system will provide recommendations and alerts, but it will not automatically restrict entry or physically manage passengers.

**Multi-Camera Synchronization:**

The current scope supports only a single mobile camera per compartment; multi-camera fusion is not implemented.

This SRS is intend**Developers:** To guide the design and implementation of the system.

- **Project Managers:** To track project scope and deliverables.

- **Testers:** To formulate test cases for verification.
- **Railway Management/Clients:** To validate that the system meets their business needs for safety and efficiency.

It is suggested to read Section 1 and 2 for an overview, followed by Section 4 Functional Requirements) and Section 5 Nonfunctional Requirements) for detailed specifications.

## 1.4 Product Scope

RailVision is a **zero-cost, vision-based crowd detection system** implemented on the Python platform. Its purpose is to enhance passenger safety and optimize carriage distribution by leveraging **existing onboard CCTV infrastructure** to process live RTSP video feeds. The core functionality includes real-time person detection, headcount tracking, and spatial occupancy estimation. The system is designed for Edge Computing deployment on existing onboard hardware.

# 2. Overall Description

## 2.1 Product Perspective

RailVision is a new, self-contained product that utilizes the train's existing CCTV system. It acts as an image processing and data analysis layer, taking the raw video stream as input and producing occupancy data as output. It does not replace any existing train systems but provides supplementary data for real-time decision-making by rail operators.

## 2.2 Product Functions

The major functions the product must perform are:

- **Video Feed Ingestion:** Connect to and ingest live RTSP video streams from onboard cameras.
- **Image Preprocessing:** Perform frame resizing, normalization, and Region of Interest (ROI) masking.
- **Person Detection:** Use YOLO models (v8/v10) to accurately detect the number of people in the frame.
- **Density Mapping:** Calculate and display the spatial occupancy percentage of the carriage.
- **Data Output:** Store and transmit real-time headcount and occupancy data.

## 2.3 User Classes and Characteristics

| User Class | Characteristics | Priorities |
|---|---|---|
| **System Administrator** | Responsible for initial setup, configuration, and maintenance. Highly technical, familiar with Linux/Edge Computing. | System health, reliability, deployment ease. |
| **Railway Operator/Manager** | Uses the system's output (occupancy data) for real-time distribution decisions and historical analysis. Non-technical, requires clear, graphical data. | Accuracy, minimal latency, clear reporting. |

## 2.4 Operating Environment

- **Deployment Platform (Edge Computing):** Existing onboard hardware (specific specifications **TBD**).
- **Operating System:** Linux (specific distribution **TBD**).
- **Software Components:** Python 3.x, OpenCV, Ultralytics YOLO (v8/v10), NumPy.
- **Input Data Source:** Live RTSP video streams from existing onboard CCTV cameras.

## 2.5 Design and Implementation Constraints

- **Zero-Cost Hardware:** Must use existing onboard CCTV cameras and computing hardware; no new hardware procurement is allowed.
- **Technology Stack:** Must use Python and open-source libraries (OpenCV, YOLO, NumPy).
- **Edge Computing:** Video processing must occur locally on the onboard hardware to ensure low latency and data privacy.
- **Data Privacy:** Video frames must not be stored externally. The system only stores and transmits numerical headcount and occupancy data.

## 2.6 User Documentation

The following user documentation components will be delivered:

- System Administrator Setup and Configuration Guide.
- API/Data Interface Specification for integration into the Rail Operations Center.

## 2.7 Assumptions and Dependencies

- **Assumption:** The existing onboard CCTV system provides a stable, accessible RTSP video feed with a resolution suitable for object detection.
- **Assumption:** The existing onboard computing hardware has sufficient CPU/GPU resources to run the YOLO model for real-time processing (i.e., at least 10 Frames Per Second).
- **Dependency:** Stable power supply and network connectivity within the train compartment for the onboard hardware.

# 3. External Interface Requirements

## 3.1 User Interfaces

A minimal local interface (potentially command-line or a simple web dashboard) is required for System Administrators for initial setup, configuration (e.g., ROI masking), and real-time monitoring of system health.

- **UI Requirement:** The system shall provide an interface for System Administrators to define the specific Region of Interest (ROI) for each camera view, masking out non-compartment areas.
- **UI Requirement:** The system shall display the calculated Headcount and Occupancy Percentage in a clear, digital format.

## 3.2 Hardware Interfaces

- **Video Input:** The system must interface with the train's network to receive the RTSP video stream from each compartment camera.
- **Processing Unit:** Direct access to the CPU/GPU resources of the onboard Edge Computing hardware for accelerated model inference.

## 3.3 Software Interfaces

- **Operating System:** Must interface with the underlying Linux OS for resource management and execution of Python scripts.
- **Deep Learning Framework:** Requires a stable interface with the Ultralytics YOLO library for model loading and inference.
- **Video Processing:** Requires integration with OpenCV for video stream capture and image manipulation.
- **Data Output API** The system shall expose a RESTful API or a similar mechanism for external systems (e.g., Central Rail Management Software) to retrieve real-time occupancy data. The data shall include:
    - TrainID
    - CarriageID

- Timestamp
- Headcount
- OccupancyPercentage

## 3.4 Communications Interfaces

- **Protocol:** Must support the **RTSP (Real-Time Streaming Protocol)** for ingesting video feeds.
- **Data Transmission:** Use **HTTP/HTTPS** for transferring final occupancy data to the Central Rail Management System.
- **Data Transfer Rate:** The final occupancy data (headcount, percentage) must be transmitted at least once every 5 seconds.

# 4. System Features

This section outlines the primary functional capabilities of the RailVision system, organized by feature.

## 4.1 Real-Time Crowd Detection

### 4.1.1 Description and Priority

This is the core functionality: processing the live video feed to determine the number of people and their spatial density. This feature is of **High** priority.

### 4.1.2 Stimulus/Response Sequences

**Sequence:** Start of Train Service

- **Stimulus:** System Administrator authorizes system startup.
- **Response:** System connects to all configured RTSP streams, loads the YOLO model, and begins the monitoring loop.

**Sequence:** Person Detection

- **Stimulus:** A new video frame is received from the camera.
- **Response:** System pre-processes the frame (ROI masking, resizing) and runs the YOLO model to generate bounding boxes for detected persons.

### 4.1.3 Functional Requirements

| ID | Functional Requirement |
|---|---|
| FR1 | **Video Source Connectivity :**<br>The system shall connect to a configurable list of RTSP stream URLs. |
| FR2 | **Image Preprocessing :**<br>The system shall preprocess video frames, including frame resizing and grayscale normalization, before feeding them to the detection model. |
| FR3 | **Area of Interest Management :**<br>The system shall apply a user-defined Region of Interest (ROI) mask to exclude non-relevant areas from the detection count. |

## 4.2 Occupancy Reporting

### 4.2.1 Description and Priority

The system must output the processed headcount and occupancy data in a usable format for external systems. This feature is of **High** priority.

| ID | Functional Requirement |
|---|---|
| FR4 | **Crowd Metrics Display :**<br>The system shall store the calculated Headcount, Occupancy Percentage, Carriage ID, and Timestamp locally for a minimum of 24 hours. |

# 5. Other Nonfunctional Requirements

## 5.1 Performance Requirements

| ID | Performance Requirement |
|---|---|

| | |
|---|---|
| NFR1 | The system must achieve a person detection and headcount frame rate of at least 10 |
| ID | Performance Requirement |
| | Frames Per Second □FPS) on the target Edge Computing hardware. |
| NFR2 | The end-to-end latency (frame capture to data output) must not exceed 500ms. |
| NFR3 | The memory footprint of the running application, including the model, must not exceed Calendar event MB of available RAM on the onboard hardware. |

## 5.2 Safety Requirements

| ID | Safety Requirement |
|---|---|
| NFR4 | The system shall be implemented to prevent any remote command execution that could interfere with or compromise the train's critical operational systems. |
| NFR5 | Failure of the crowd detection system must not impact the operation or stability of the CCTV feed or the train's network infrastructure. |

## 5.3 Security Requirements

| ID | Security Requirement |
|---|---|

| | |
|---|---|
| NFR6 | The system must ensure data privacy by processing video locally Edge Computing) and not transmitting or storing raw video frames or images externally. |
| NFR7 | Access to the system's configuration interface and API endpoints must be |
| ID | Security Requirement |
| | protected by robust user authentication (e.g., API keys or OAuth). |
| NFR8 | All data transmission from the onboard unit to external systems must be encrypted (e.g., HTTPS. |

## 5.4 Software Quality Attributes

- **Accuracy:** The person detection accuracy F1Score) must be maintained at 95% or higher under varying lighting conditions.
- **Robustness:** The system must be able to recover automatically from temporary loss of the RTSP video feed without requiring a manual restart.
- **Maintainability:** The codebase must be modular, adhering to Python best practices, and the deep learning model should be easily updatable with new versions of YOLO (e.g., v10) without major system redesign.

## 5.5 Business Rules

- **Data Use:** Occupancy data will only be used for carriage management, operational reporting, and safety analysis, not for individual passenger tracking.
- **System Status:** The system must clearly indicate if a camera feed is offline or if the processing performance is degraded.

# 6. Other Requirements

- **Internationalization:** The system must be designed to be adaptable to different carriage layouts and ROI configurations.

- **Database Requirements:** A local time-series database (e.g., SQLite or InfluxDB should be used on the edge device for temporary storage FR8.

# Appendix A Glossary

| Term | Definition |
|---|---|
| **RTSP** | Real-Time Streaming Protocol, used to stream video feeds over a network. |
| **ROI** | Region of Interest, the specific area within the video frame where detection is performed. |
| **YOLO** | You Only Look Once, a state-of-the-art real-time object detection model. |
| **Edge Computing** | Data processing is done on the local device (the train's onboard hardware) rather than in a centralized cloud or server. |
| **Headcount** | The numerical count of detected persons in the ROI. |
| **Occupancy Percentage** | The estimated spatial density of the carriage, calculated from person count and ROI area. |

# Appendix B Analysis Models

A high-level data flow diagram showing RTSP Ingestion → Preprocessing → YOLO Model → Headcount/Density Map → API Output]

# Appendix C To Be Determined List

| ID | Section | Description |
|---|---|---|

| | | |
|---|---|---|
| TBD1 | 2.4 | Specific CPU/GPU specifications of the existing onboard hardware. |
| TBD2 | 5.1 | Exact MB limit for system memory footprint NFR3. |

| ID | Section | Description |
|---|---|---|
| TBD3 | 4.1.2 | Specific action the system takes if the YOLO model fails to load. |